

DECA Section 3

2025年11月11日 星期二

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DECA S3.

7 November, 2025

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Reference DECA section 3.

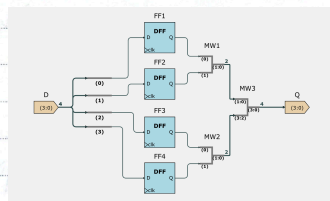
Aim: Learn about D-type flip flop, which is a unit of information storage.

It stores input D when clock input change from low $\rightarrow$ high

Try to create counter & countdown logic.

Procedure.

- Build a DFF with 4 bits, in which all 4 bits share a same clock input.



- We can simulate "DFF4" with step simulation
Which works successfully

- Counter

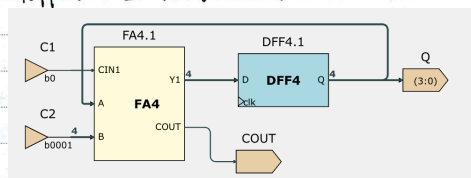
Then we can build a sequential component, where the next input D_{n+1} depends on current output Q_n .

Initially, A is at default number of 0000, B is constant 0001

Each time they make a full-adder, which is $A+1$.

When clock rises, D passes to Q and Q passes back to A, a new "add one" happens, so it counts one.

(increment)



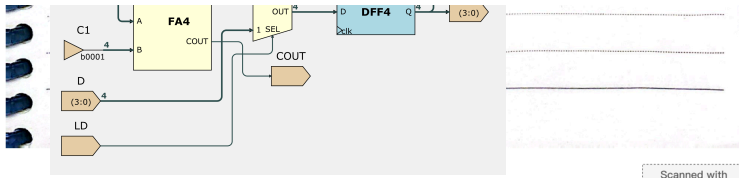
When $Q = 1111$, in FA4: $1111 + 0001 = 0000 + 1$ in COUT.
So it comes back to 0 and counts again.

- Counter with synchronous load

By adding a MUX, we can input a start of counter we want by putting it into input B, set LD to 1.

Next, we set LD back to 0 then it starts counting.





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• Challenge: Countdown timer with auto-reset.

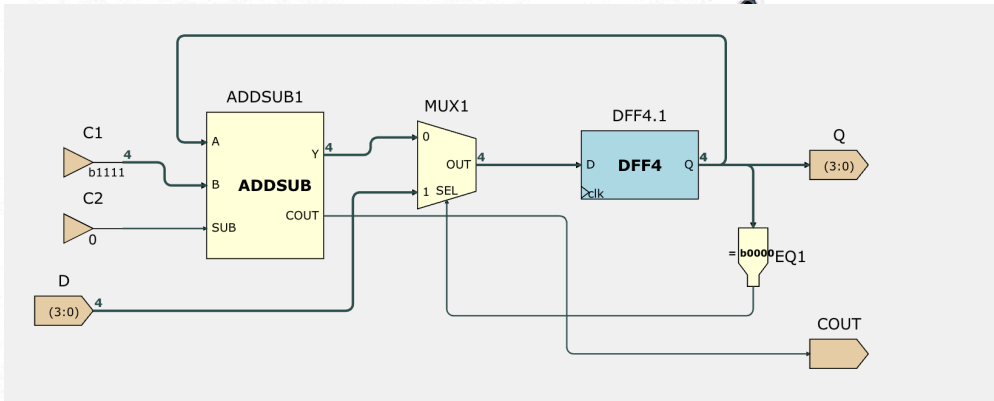
First, replace FA4 with "add-sub" and set $C_{in}=1$, so it's subtractor.

Second, we connect a reduction NOR gate with SELECT in MUX. (Eq. 1)

Eq. 1 will give 1 in initial condition as default input is 'b0000', so output is 1. Then MUX will choose and input D, the countdown runs.

When $Q \neq b0000$, output of Eq. 1 will never be 1, so counter will countdown continuously: to 0

At $Q=0$, counter loads for D+1 times. Eq. 1 gives 1, so the counter goes back to D and repeats.



Ref.